CHEST DISEASES HOSPITAL | LUNG TRANSPLANT UNIT

Adult Lung Transplant Anaesthesia Protocol

Multidisciplinary Perioperative Anaesthesia Protocol

Version 1.0
April 2026
Ref: CDH-LTU-ANAES-01

Document Scope	Intraoperative anaesthesia management for adult lung transplantation
Patient Population	Adult candidates (≥18 years) for single or bilateral lung transplantation
Applicable Centre	Lung Transplant Unit, Chest Diseases Hospital, Ministry of Health, Kuwait
Related Protocols	CDH-LTU-PERIOP-02 (Perioperative Protocol) • CDH-LTU-PPO-03 (Post-operative Orders) • CDH-LTU-IMPL-01 (LTU Implementation Protocol)

CROSS-REFERENCE This protocol is to be read in conjunction with the CDH LTU Perioperative Protocol (CDH-LTU-PERIOP-02) and the Post-operative Orders (CDH-LTU-PPO-03). Anaesthesia decisions remain the responsibility of the Lead Transplant Anaesthesiologist at all times.

SECTION 1: OVERVIEW & TEAM ROLES

1.1 Purpose and Scope

This protocol defines the step-by-step intraoperative anaesthetic management of adult patients undergoing lung transplantation at Chest Diseases Hospital. It covers all team members involved in anaesthesia care from pre-operative assessment through to safe transfer to the ICU. Lung transplantation requires a highly coordinated multidisciplinary approach — all team members must be familiar with their specific duties as outlined below.

1.2 Key Team Members — Anaesthesia Domain

Role	Duties & Authority
Lead Transplant Anaesthesiologist	Duty Consultant/Specialist — final clinical authority for all anaesthesia decisions
Second Anaesthesiologist	Consultant/Specialist or Senior Registrar — two anaesthetists required at all lung transplant procedures
Anaesthesia Assistant	Prepares equipment, assists induction, manages airway adjuncts, circulates drugs, positions defibrillator pads
ECMO Specialist / Perfusionist	Senior perfusionist present from induction until ICU transfer; prepares and manages CPB/ECMO circuit; operates cell salvage equipment
Transplant Surgeon	Present at induction; communicates timing for cross-clamp, reperfusion, and ECMO activation
Scrub Nurse / Circulating Nurse	Prepares surgical equipment; assists with ECMO circuit preparation; manages cell salvage
Transplant Coordinator	Coordinates timing between retrieval team and OR; communicates cold ischaemia time to anaesthesia team

SECTION 2: PRE-OPERATIVE PHASE

2.1 Lead Transplant Anaesthesiologist — Pre-operative Duties

Prior to day of surgery:

- Assess aetiology and severity of underlying lung disease (pulmonary fibrosis, COPD, cystic fibrosis, pulmonary hypertension). Review baseline ABG and PFTs.
- Review most recent 6MWT, echocardiogram, right heart catheterisation data, and CT chest.
- Assess for significant pulmonary hypertension: resting mean PAP >25 mmHg indicates high risk for induction.
- Obtain data regarding previous anaesthetic exposures, surgeries, drug sensitivities, laboratory values, and other organ function.

- Evaluate and ensure preoperative treatment of active pulmonary infections.
- Nutritional assessment; correct electrolyte disturbances.
- Review all current medications: PDE-5 inhibitors, prostanoids, ERA (PAH); bronchodilators; immunosuppressants.

On day of surgery:

- Obtain donor details from transplant physician: age, weight, height, cause of death, ventilatory parameters, FiO_2, fluid balance, haemodynamic support, infection parameters, and antibiotics administered.
- Confirm cold ischaemia time with transplant coordinator; brief entire anaesthesia team on urgency.
- Continue pulmonary vasodilators preoperatively (sildenafil, inhaled prostacyclin).
- Administer antibiotic prophylaxis per institutional protocol (Meropenem + Vancomycin if no specific directions received).
- Confirm tacrolimus/cyclosporin dose administered before OR transfer (per transplant physician instruction).
- Identify planned surgical approach: bilateral sequential thoracotomy (clamshell), sternotomy, or VATS-assisted.
- Determine ECMO strategy with transplant surgeon: if PA clamping intolerance is likely (PAH, severe RV dysfunction), plan pre-induction ECMO.
- Check blood products availability and clotting status; alert haematology if TEG/ROTEM-guided transfusion anticipated.
- Discuss defibrillator pad placement with Anaesthesia Technician — pads must be placed before draping.

PAH CRITICAL ALERT PAH PATIENTS: Awake arterial line and awake femoral cannulation for ECMO may be required BEFORE induction. Positive pressure ventilation, repositioning, and vasodilating anaesthetic drugs are critical high-risk waypoints in PAH patients — have vasopressors prepared before every step.

2.2 Anaesthesia Assistant — Equipment Preparation

Anaesthesia machine check:

- Ventilator function and gas supplies confirmed.
- Calibrate gas analyser and capnograph — $ETCO_2$ is a critical monitor in ECMO cases.
- Two functional suction devices prepared.

Draw up and label all induction drugs:

- Induction agents: Propofol, Ketamine, Midazolam, Etomidate (PAH/high-risk back-up).
- Opioids: Fentanyl/Sufentanil (depending on availability).
- Muscle relaxant: Rocuronium/Vecuronium/Cisatracurium.
- Emergency vasopressors: epinephrine (diluted), norepinephrine, vasopressin (diluted).
- Rapid bolus: metaraminol or phenylephrine.
- Calcium chloride 10% prepared for reperfusion.
- Methylprednisolone prepared per institutional protocol (see Section 8).
- Induction immunosuppression: Alemtuzumab 30 mg IV or Basiliximab 20 mg IV — as directed.

Airway equipment:

- Left-sided double-lumen tube (DLT): 37F/39F adult female; 39F/41F adult male.
- Single-lumen ETT for CPB cases or end-of-procedure exchange.
- Video laryngoscope (C-MAC/GlideScope) AND standard laryngoscope — both checked.
- Fibreoptic bronchoscope connected, light source and image confirmed.
- Cook exchange catheter (airway exchange catheter) for DLT-to-SLT exchange at end of case.
- Difficult airway cart confirmed available.

Additional preparation:

- Level 1 rapid infuser (or equivalent) primed; TEG/ROTEM analyser available
- Prepare UFH bolus dose: 2000–5000 IU flat dose (or 25–100 IU/kg).
- Confirm drug storage refrigerator contains immunosuppressants with correct pharmacy labels.
- Place defibrillator pads per surgeon instruction before draping.

2.3 Perfusionist — CPB/ECMO Circuit Preparation

- Confirm ECMO vs CPB strategy with transplant surgeon before patient arrives.
- Prime CPB/ECMO circuit per institutional protocol — have circuit ready before induction in high-risk cases.

- Confirm cannula sizes with surgeon (central vs peripheral strategy).
- Calibrate ACT monitor: target ACT 180–220 seconds during ECMO; >480 seconds for CPB.
- Confirm sweep gas supply (O_2 and air blender) for ECMO oxygenator.
- Set up cell salvage circuit from skin incision.
- Remain present and on standby until patient is safely transferred to ICU.

2.4 Transplant Coordinator — Pre-operative Coordination

- Confirm retrieval team ETA and cold ischaemia clock to OR team; relay to lead anaesthesiologist.
- Confirm donor lung quality report (PaO_2/FiO_2 ratio, bronchoscopy findings) is available for anaesthesiologist review.
- Relay any last-minute changes in organ quality or retrieval complications to surgical and anaesthesia teams immediately.
- Ensure recipient consent and pre-operative checklist are complete.

SECTION 3: MONITORING SETUP

MONITORING STANDARD STRICT ASEPTIC PRECAUTIONS must be followed during ALL invasive monitoring procedures. Standard ASA monitoring is the minimum; advanced monitoring as below is mandatory for all lung transplant cases.

Monitor	Specification / Notes
Continuous ECG	ST-segment analysis throughout. 5-lead ECG minimum.
Arterial Line	RIGHT radial artery preferred — monitors for North-South syndrome if peripheral VA-ECMO used.
Pulse Oximetry	2 probes (both hands); right finger pulse oximeter as peripheral perfusion marker alongside arterial line.
Peripheral IV Access	Minimum two large-bore peripheral IV cannulas (≥16G) for rapid volume administration.
Central Venous Catheter (CVC)	Preferred site: right internal jugular vein (IJV) + vascular sheath introducer. PROTECT right IJV for VV-ECMO access if post-operative ECMO anticipated. Alternative: left IJV or subclavian.
Pulmonary Artery Catheter (PAC)	Left-sided insertion preferred. Advance to proximal PA and confirm on TEE BEFORE pneumonectomy. Withdraw PAC to proximal PA/RVOT before pneumonectomy to prevent PA injury.
TEE (Transoesophageal Echo)	Insert immediately after intubation. Mandatory for all cases. See Section 3.1.
BIS Monitor	Mandatory during VA-ECMO — volatile agents not taken up via lungs on bypass; TIVA in use. Target BIS 40–60.
Cerebral Oximetry (NIRS)	Near-infrared spectroscopy; marker of cerebral oxygenation during peripheral ECMO.
Temperature Probe	Oesophageal or nasopharyngeal.
$ETCO_2$	Critical qualitative indicator: $ETCO_2$ >20 mmHg confirms transpulmonary blood flow on VA-ECMO.

3.1 TEE Surveillance Protocol

Operative Phase	TEE Mandate
Post-intubation (baseline)	Full baseline biventricular assessment, intracardiac shunts, PA/PV anatomy.
Before pneumonectomy	Guide PAC withdrawal to proximal PA/RVOT. Confirm PA catheter position.
During PA clamping	Monitor for RV dilatation, septal deviation, haemodynamic response — alert for RV failure.
On VA-ECMO	Monitor LV ejection; detect LV overdistension (domed septum); guide LV venting if required.

Post-PA anastomosis	Colour Doppler: PA anastomotic diameter ≥75% of donor PA. Alert surgeon if <75%.
Post-PV anastomosis	PV diameter <2.5 mm or velocity >1.6 m/s indicates anastomotic stenosis — alert surgeon immediately.
De-airing phase	Monitor for air in LA, LV, and coronary ostia during final anastomosis stages.
ECMO removal	Confirm adequate biventricular function and lung graft function before decannulation.

SECTION 4: ANAESTHETIC INDUCTION

4.1 Standard Induction — Low-to-Moderate Risk (Non-PAH)

Goals: maintain haemodynamic stability; avoid hypotension; ensure adequate oxygenation; minimise PVR increases.

- Preoxygenation: 100% FiO_2 to prevent desaturation during induction.
- Position: semi-sitting as per patient comfort. Treat as full stomach — use rapid sequence induction.
- Induction agents: Propofol (if haemodynamically stable, avoid large boluses) OR Etomidate/Ketamine (unstable patients).
- Anxiolysis: Midazolam 1–2 mg IV as adjunct if haemodynamics permit.
- Opioids: Sufentanil or Fentanyl (per availability) to blunt sympathetic response to laryngoscopy. Maintain infusion for adequate intraoperative analgesia.
- Neuromuscular blockade: Rocuronium (preferred) or Vecuronium for optimal intubation conditions.
- Advance left-sided DLT under direct vision; confirm position with fibreoptic bronchoscope immediately after intubation.

CRITICAL If haemodynamic instability occurs at induction: start epinephrine infusion pre-emptively for severe RV dysfunction. Have diluted epinephrine prepared before induction commences.

4.2 High-Risk / PAH Induction Protocol

- Discuss pre-induction ECMO with surgeon — consider awake femoral cannulation before induction.
- Limit premedication: Midazolam 1–2 mg IV only if respiratory and haemodynamic status permits.
- Primary induction agent: Etomidate 0.3 mg/kg IV — minimal myocardial depression.
- Maintain vasopressor infusion (epinephrine or norepinephrine) running from before induction commences.
- Avoid agents that reduce systemic vascular resistance precipitously (high-dose propofol bolus).
- Confirm ECMO circuit primed and perfusionist standing by before commencing induction.

⚠ **WARNING** HIGH-RISK WAYPOINTS in PAH: (1) Initiation of positive pressure ventilation. (2) Lateral repositioning. (3) Drugs causing afterload reduction. Have vasopressors immediately available at each waypoint. (Murray JTD 2021)

4.3 Disease-Specific Induction Considerations

Diagnosis	Specific Considerations
COPD / Emphysema	Risk of air trapping causing hypotension on PPV. Prolonged expiratory time (I:E 1:3 to 1:4). Avoid auto-PEEP. Consider haemodynamic collapse from air-trapping as cause of instability.
ILD / IPF (Restrictive)	Elevated peak inspiratory pressures from decreased compliance. Anticipate need for VA-ECMO. Careful fluid management is critical.
Pulmonary Arterial Hypertension	Pre-induction ECMO may be required. Awake arterial line mandatory. High-risk waypoints: PPV, repositioning, vasodilating drugs. Serial echo for RV remodelling post-operatively.
Cystic Fibrosis	Commence with single-lumen ETT; perform therapeutic bronchoscopy first; then exchange to DLT. Cross-infection precautions.

4.4 Anaesthesia Assistant — Induction Support

- Confirm all drugs drawn up and labelled before patient arrives in OR.

- Assist laryngoscopy — apply cricoid pressure if requested by lead anaesthesiologist.
- Hand DLT to anaesthesiologist; ensure fibreoptic bronchoscope is ready for immediate confirmation.
- Connect DLT to circuit; confirm bilateral breath sounds and $ETCO_2$ trace.
- Prepare TEE probe; assist anaesthesiologist for insertion immediately post-intubation.
- Connect Level 1 rapid infuser to large-bore IV access as directed.

SECTION 5: AIRWAY MANAGEMENT

5.1 Double-Lumen Tube (DLT) — Primary Approach

- Select LEFT-sided DLT for the majority of cases, including off-pump and ECMO cases.
- Size selection: 37F/39F for adult female; 39F/41F for adult male.
- Confirm DLT position with fibreoptic bronchoscope after intubation AND after patient repositioning.
- Re-confirm DLT position before each OLV phase.
- Bronchial blockers may be used if DLT placement is difficult.

⚠ **WARNING** Re-confirm DLT position with fibreoptic bronchoscopy after patient is repositioned to lateral decubitus — tube migration is common with positioning changes.

5.2 Single-Lumen ETT — Specific Indications

- Use single-lumen ETT if cardiopulmonary bypass (CPB) is the planned intraoperative support strategy.
- Cystic Fibrosis: commence with single-lumen ETT → therapeutic bronchoscopy → then exchange to DLT.
- Difficult airway: secure airway with SLT first, then exchange to DLT once stable.

5.3 DLT to Single-Lumen ETT Exchange — End of Procedure

- Plan DLT-to-SLT exchange at end of bilateral procedure.
- Use Cook airway exchange catheter through the DLT lumen.
- Use video laryngoscope to visualise glottis during exchange.
- Insert appropriate single-lumen cuffed ETT over exchange catheter under direct vision.
- Confirm position with $ETCO_2$ and bilateral breath sounds.
- Perform bronchoscopy before ICU transfer — inspect anastomoses and clear secretions.
- CHECK cuff pressure — AVOID tracheal ischaemia with excessive cuff pressures. Tolerate mild leaks.

SECTION 6: ANAESTHESIA MAINTENANCE & ONE-LUNG VENTILATION

6.1 Anaesthetic Agents

Mode	Protocol
Inhalational (off-ECMO)	Sevoflurane titrated to adequate depth — confirm with BIS target 40–60. Supplement with opioid infusion (Sufentanil/Fentanyl). Maintain neuromuscular blockade as required.
TIVA on VA-ECMO — MANDATORY	Switch to TIVA immediately when VA-ECMO is initiated. Volatile agents are NOT taken up by the lungs during bypass. Propofol infusion titrated to BIS 40–60. Opioid infusion continued. Do NOT rely on volatile agent monitoring during ECMO.

6.2 One-Lung Ventilation (OLV) Settings — Disease-Specific

PROTOCOL NOTE Ventilation strategy is DISEASE-SPECIFIC. Apply the correct settings based on recipient diagnosis and operative phase. Lung protection is the priority after reperfusion.

Phase / Diagnosis	Ventilator Settings
COPD/Obstructive (native lung OLV)	I:E 1:3 to 1:4 | RR 10–12/min | Minimal/no PEEP (auto-PEEP risk) | Permissive hypercapnia acceptable | Watch for haemodynamic collapse from air-trapping
ILD/Restrictive (native lung OLV)	I:E 1:1 to 1.5:1 | RR 14–18/min | TV 4–5 mL/kg IBW | PEEP 8–12 cmH_2O | Expect high Pplat from low compliance

General OLV — pre-transplant	Standard OLV to maintain gas exchange and haemodynamic stability. Priority: gas exchange and circulation. Lung-protection secondary at this phase.
Post-reperfusion — first lung	TV 3–4 mL/kg DONOR IBW | FiO_2 <0.4 | PEEP 5–8 cmH_2O | Dual-mode ventilation | Lung-protective strategy applies from reperfusion
Post-reperfusion — both lungs	TV 6–8 mL/kg DONOR IBW | FiO_2 lowest to maintain SpO_2 ≥92% | PEEP 5–8 cmH_2O
FiO_2 at moment of reperfusion	FiO_2 0.21 (room air) at moment of reperfusion — lowest possible. Avoid hyperoxia at all costs. Titrate to maintain SpO_2 >94%.

6.3 Intraoperative Haemodynamic Targets

Parameter	Target
MAP (on ECMO)	60–65 mmHg (maintain coronary perfusion)
PA Systolic Pressure (intraop)	<40 mmHg during surgery; target 10–15 mmHg after second lung implanted
Cardiac Index (ICU)	1.8–2.2 L/min/m² (avoid higher CI)
CVP (post-op target)	<7 mmHg
Haemoglobin	Target Hb >9 g/dL. Use TEG/ROTEM to guide blood product administration.
ECMO Pulsatility	Maintain pulsatility in systemic and PA waveforms during VA-ECMO.
Total Intraoperative Fluids	≤2000 mL total. Crystalloid + colloid (albumin) + blood products guided by TEG/ROTEM.

⚠ **WARNING** AVOID NSAIDs throughout the entire perioperative period — renal risk is amplified by concurrent calcineurin inhibitors (tacrolimus/cyclosporin).

6.4 Additional Maintenance Duties

- Administer prophylactic antibiotics 30 minutes before skin incision, adjusted per recipient airway culture/colonisation profile.
- Monitor BIS continuously for depth of anaesthesia.
- Monitor blood loss; TEG/ROTEM results; prepare blood products as directed.
- Activate cell salvage from time of skin incision.
- Maintain warming device (forced-air blanket) — prevent perioperative hypothermia.
- Record hourly urine output — alert lead anaesthesiologist if <0.5 mL/kg/h.
- Administer induction immunosuppression as directed: Alemtuzumab 30 mg IV OR Basiliximab 20 mg IV via dedicated IV line; document time and drug batch number.

SECTION 7: ECMO / ECLS MANAGEMENT

📌 **NOTE** VA-ECMO is preferred over CPB for intraoperative support. Lower heparin requirements and reduced systemic inflammatory response.

7.1 Timing and Indication

- Pre-induction ECMO: mandated for severe cardiopulmonary failure (severe PAH, severe RV dysfunction, low cardiac reserve).
- Rescue ECMO: activate immediately for haemodynamic collapse on PA clamping, RV failure, or severe desaturation refractory to conventional management.
- PA clamping test: clamp PA for 5–10 minutes pre-pneumonectomy while monitoring TEE for RV dilatation and haemodynamic tolerance. If intolerant → activate ECMO before pneumonectomy.

7.2 Cannulation Strategy

Strategy	Indication & Details
Central V-A ECMO (RA→Aorta)	Preferred for intraoperative support. Direct cannulation by surgeon.

Peripheral V-A ECMO (Femoral)	Preferred for post-operative continuation or rescue. Femoral vessels prepped and exposed before incision if anticipated.
VV-ECMO	Consider for isolated respiratory failure without cardiac compromise. Protect right IJV for VV-ECMO access.

7.3 Heparin and Anticoagulation

- Administer UFH after surgical dissection is complete: flat dose 2000–5000 IU IV. Low-dose protocol acceptable if ECMO flows are maintained.
- Target ACT 180–220 seconds during ECMO — check every 30–60 minutes.
- Target ACT >480 seconds for CPB.

7.4 ECMO Flow Management

- Start at 50% of calculated cardiac output — adjust per haemodynamics and gas exchange.
- Maintain pulsatility in systemic and PA waveforms during VA-ECMO.
- Monitor for LV overdistension during VA-ECMO — confirm LV ejection on TEE.
- Monitor $ETCO_2$ continuously: $ETCO_2$ >20 mmHg confirms transpulmonary blood flow.
- Perfusionist: adjust sweep gas (FiO_2 and gas flow) to target pCO_2 and PaO_2 per lead anaesthesiologist direction.

7.5 ECMO Weaning

- Reduce ECMO flow to 1 L/min and hold 10–20 minutes before terminating.
- Assess biventricular function and graft function on TEE before decannulation.
- Prolong ECMO if: high ventilatory pressures, decreased graft compliance, or severe RV dysfunction persists.
- Perform final de-airing under TEE guidance (pulmonary veins, LA, LV) before ECMO removal.

SECTION 8: REPERFUSION PROTOCOL

CRITICAL ALERT Reperfusion is the most critical haemodynamic waypoint. Have all vasopressors and calcium chloride immediately available BEFORE PA clamp release. Communicate clamp release timing clearly between surgeon and anaesthesiologist.

8.1 Pre-reperfusion Checklist

- Administer Methylprednisolone 500 mg IV before reperfusion — document dose and time (ICU team to schedule three subsequent doses).
- Perform bronchoscopy through DLT BEFORE pulmonary artery clamp release: inspect bronchial anastomosis integrity and mucosal viability; clear secretions from donor bronchi.
- Set FiO_2 to 0.21 (room air) at moment of reperfusion — lowest possible.
- Increase vasopressors BEFORE clamp release — prepare for acute hypotension.
- Calcium chloride 10% IV bolus: prepare syringe labelled and ready at reperfusion.
- Epinephrine diluted bolus syringes: ready for rapid administration.

8.2 Reperfusion Sequence

- Release PA clamp slowly and gradually — not abruptly.
- Haemodynamic support: IV fluid boluses (within restrictive target); Epinephrine 5–10 µg IV boluses for instability; Calcium chloride 10% IV bolus at reperfusion.
- Monitor $ETCO_2$ after clamp release — rising $ETCO_2$ >20 mmHg confirms transpulmonary blood flow.
- After reperfusion: switch to post-reperfusion protective ventilation settings (TV 3–4 mL/kg DONOR IBW; FiO_2 0.3–0.4; PEEP 5–10 cmH_2O).
- Reassess TEE immediately: graft function, PA and PV anastomotic patency, RV response, LV function.

8.3 Primary Graft Dysfunction (PGD) — Intraoperative Recognition

ISHLT DEFINITION ISHLT PGD Definition: New bilateral infiltrates on CXR/imaging with a deteriorating PaO_2/FiO_2 ratio within 72 hours of transplant, after excluding other causes (anastomotic complication, haemothorax, lobar torsion, cardiogenic oedema).

PGD Grade	PaO_2/FiO_2 Ratio	Anaesthesia Action
Grade 0	≥300 mmHg	Routine lung-protective ventilation; monitor graft function closely.
Grade 1	200–299 mmHg	Intensify lung-protective ventilation; review fluid balance; monitor hourly.
Grade 2	100–199 mmHg	Increase PEEP (8–12 cmH_2O); Pplat <30 cmH_2O; consider or continue ECMO; alert surgical team.
Grade 3	<100 mmHg — Severe PGD	Mandatory lung-protective ventilation (TV 4–6 mL/kg IBW; PEEP 8–12 cmH_2O). Initiate or continue VA-ECMO/VV-ECMO. Minimise FiO_2. Communicate grade to ICU team.

SECTION 9: ANALGESIA PROTOCOLS

Incision Type	Preferred Regional Technique	Alternative
Bilateral thoracotomy	Thoracic Epidural (TEA) T4–T6	Bilateral ESPB if anticoagulated
Clamshell incision	Thoracic Epidural T4–T6	Bilateral ESPB or bilateral TPVB
Sternotomy	Thoracic Epidural T4–T6	Bilateral ESPB or bilateral TPVB
Anticoagulated patient	Bilateral ESPB (continuous catheter) — SAFE with anticoagulation; fascial plane block	Bilateral TPVB if ESPB not feasible

9.1 Thoracic Epidural Anaesthesia (TEA)

- TEA is the preferred analgesic modality after lung transplantation: reduces postoperative mechanical ventilation time and improves pulmonary rehabilitation.
- Level: T4–T6 interspace for bilateral thoracotomy or clamshell incision.
- Placement timing: if significant intraoperative anticoagulation was used, place thoracic epidural POST-operatively after anticoagulation has cleared, to avoid spinal haematoma.

9.2 Erector Spinae Plane Block (ESPB) — Anticoagulated Patients

- ESPB is SAFE with anticoagulation — fascial plane block (not neuraxial).
- Continuous bilateral ESPB catheter technique: Level T5 bilaterally (5th thoracic transverse process). Place under ultrasound guidance.
- ESPB and TPVB are equivalent for analgesia beyond 6 hours post-operatively.

9.3 Multimodal Analgesia — All Patients

- Paracetamol 1 g IV every 6 hours (scheduled).
- Opioid infusion (Remifentanil or Fentanyl) — lowest effective dose.
- AVOID NSAIDs: renal risk amplified by concurrent calcineurin inhibitors (tacrolimus/cyclosporin). NSAIDs are CONTRAINDICATED throughout the perioperative period.

SECTION 10: IMMUNOSUPPRESSION — ANAESTHESIA ROLE

10.1 Lead Anaesthesiologist — Immunosuppression Responsibilities

- Confirm tacrolimus or cyclosporin dose administered before OR transfer (per transplant physician instruction) — document in anaesthesia chart.
- Administer induction immunosuppression in OR per institutional protocol:
  - Alemtuzumab 30 mg IV (monoclonal antibody induction): administer via dedicated IV line; pre-medicate as per protocol; have antihistamine and hydrocortisone available for infusion reactions.
  - OR Basiliximab 20 mg IV (IL-2 receptor antagonist induction): administer on POD 0 and POD 4.

- Administer via dedicated IV line — document time of administration and drug batch in anaesthesia record.
- Administer Methylprednisolone 500 mg IV before reperfusion of lungs (see Section 8).

10.2 Anaesthesia Assistant — Immunosuppression Support

- Confirm immunosuppressant drugs stored per manufacturer temperature requirements in designated refrigerator.
- Check pharmacy label, drug name, dose, batch number, and expiry date before administering.

NOTE Dedicated facilities for immunosuppressive drug storage, administration, and monitoring must be present in the anaesthetic room and operating theatre at all times.

SECTION 11: END OF PROCEDURE & TRANSFER TO ICU

11.1 Lead Anaesthesiologist — End of Case Duties

- Perform final bronchoscopy before transfer: inspect both bronchial anastomoses; document mucosal viability; clear secretions.
- Exchange DLT to single-lumen ETT (see Section 5.3).
- Confirm final TEE assessment: biventricular function, graft function, anastomotic patency.
- Confirm ECMO decannulation is complete and haemostasis achieved before transfer.
- Optimise ventilator settings for transport: TV 6–8 mL/kg IBW; lowest FiO_2 maintaining SpO_2 >94%; PEEP 5–8 cmH_2O.
- Confirm transport ventilator parameters match OR ventilator settings.
- Confirm haemodynamic stability before departure from OR.
- Accompany patient to ICU with adequate transport monitoring (SpO_2, $ETCO_2$, arterial line, ECG).
- Complete formal ICU handover to receiving ICU consultant and intensivist.

11.2 Formal Handover Checklist — ICU Team

	Handover Item
☐	Operative summary: approach (bilateral/single; clamshell/sternotomy); ECMO used (Y/N); duration
☐	Cold ischaemia times: first and second lung
☐	Intraoperative events: haemodynamic instability, ECMO complications, arrhythmias
☐	Fluid balance: total input, output, blood products administered
☐	Immunosuppression: agents used, doses, times administered
☐	Methylprednisolone: dose given and exact time — ICU team to schedule subsequent doses
☐	Analgesic plan: epidural or ESPB in situ, current drug and infusion rate
☐	Antibiotics: agent, dose, time given
☐	Current vasopressor requirements and doses
☐	Ventilator settings at time of transfer
☐	PGD Grade at time of transfer (if assessed)

SECTION 12: EMERGENCY PROTOCOLS

EMERGENCY STANDARD All emergency scenarios below require immediate communication between lead anaesthesiologist, transplant surgeon, and perfusionist. Document time of event, interventions, and responses in real time.

12.1 Haemodynamic Collapse on PA Clamping

- Immediately increase vasopressor infusion.

- Administer epinephrine IV boluses (5–10 µg).
- Request ECMO specialist to initiate ECMO — do not delay.
- Alert surgeon: PA clamp may need to be released while ECMO is being established.
- Document time of event and all interventions.

12.2 Intraoperative RV Failure

- Establish VA-ECMO if not already running.
- Administer epinephrine infusion — titrate to haemodynamic response.
- Optimise ventilation to reduce RV afterload: avoid hypoxia, hypercarbia, acidosis, and overinflation.
- Assess on TEE: RV dilatation, septal position, LV underfilling.
- Milrinone may be added per surgical/cardiology direction.

12.3 Severe Desaturation During OLV

- Apply CPAP 5–10 cmH_2O to non-ventilated (operated) lung if surgically feasible.
- Apply PEEP to ventilated (dependent) lung.
- Increase FiO_2 to 1.0 temporarily.
- Request surgeon to briefly limit lung retraction.
- If refractory: activate ECMO.

12.4 Primary Graft Dysfunction (PGD) Signs — Intraoperative

- Confirm diagnosis: new bilateral infiltrates + deteriorating PaO_2/FiO_2 ratio after reperfusion.
- Implement lung-protective ventilation: TV 4–6 mL/kg IBW; PEEP 8–12 cmH_2O; Pplat <30 cmH_2O.
- Minimise FiO_2 (target SpO_2 >92% with lowest FiO_2 possible).
- Maintain restrictive fluid strategy.
- If refractory hypoxaemia: continue or reinitiate VA-ECMO/VV-ECMO.
- Document PGD grade and communicate to ICU team at handover (see Section 8.3 for grading).

12.5 Anaphylaxis / Infusion Reaction (Immunosuppression)

- Stop the offending infusion immediately.
- Administer adrenaline 0.1–0.5 mg IV (or IM if IV not immediately available).
- Administer hydrocortisone 200 mg IV + chlorphenamine 10 mg IV.
- Support airway and ventilation.
- Initiate fluid resuscitation.
- Document reaction and alert transplant team and pharmacy.

SECTION 13: DOCUMENT CONTROL

Version	Details
1.0	Initial release — April 2026. Chest Diseases Hospital, LTU.
Review Due	April 2027 or following any critical incident or protocol deviation.
Approved By	Head of Lung Transplant Unit, CDH / Director of Anaesthesia, CDH